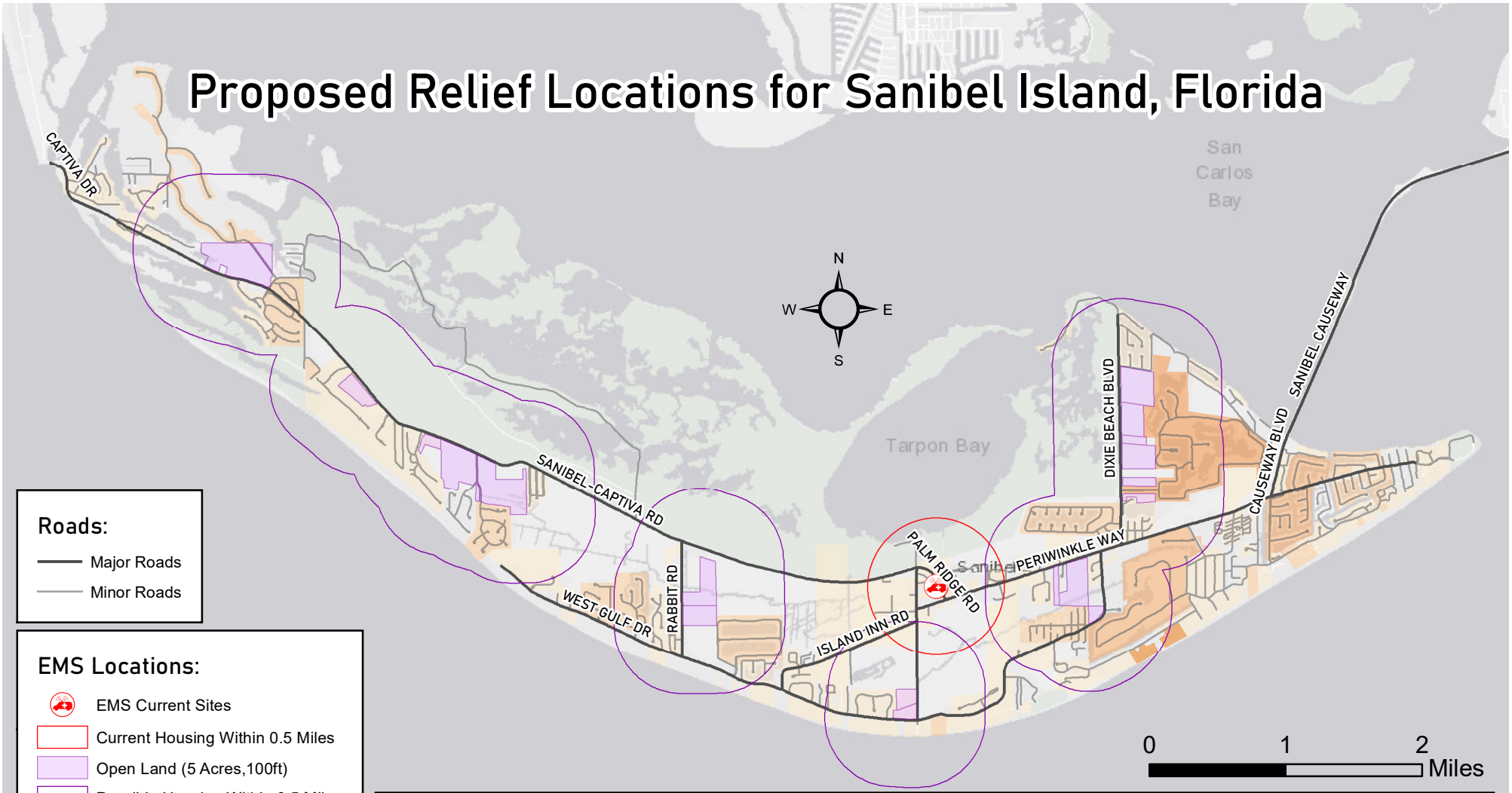
Proposed Relief Locations for Sanibel Island, Florida



Roads:

- Major Roads
- Minor Roads

EMS Locations:

- EMS Current Sites
- Current Housing Within 0.5 Miles
- Open Land (5 Acres,100ft)
- Possible Housing Within 0.5 Miles

Residential Developments:

Number of Residential Parcels

- 0 - 100
- 101 - 200
- 201 - 300
- 301 - 400
- 401 - 500

Methods:

Based on Sanibel Island's local housing distribution, and major road locations, where should a new health care (EMS or hospital) location be placed to provide relief to residents? Current roadways, housing communities, land use, and EMS locations, as provided by the Lee County GIS Database, were applied to Ersi's "Light Grey" basemap. (A separate digitized boundary polygon was made in order to clip out data that was specific to Sanibel Island from Lee County's files to reduce computer lag.) Firstly, housing communities were "symbolized" by the amount of residential parcels they contained; this shows what communities have the highest density of residents and may require higher rates of relief. A layer depicting all land use for Sanibel received a "definition query" (PLUC = VA) to single out all vacant land on the island. Furthermore, a secondary "query" was applied to single out areas that were over five acres large (Acres > 5). No functions were performed on minor roadways, however, major roadways received a 100 foot "buffer"; this "buffer" showed which vacant land was within 100 feet of major roads, possibly for emergency relief travel. An "intersection" between the 100 foot "buffer" and the five acre vacant land areas showed 17 sites directly intersected major roadways. These sites were singled out through a third "definition query" that singled out the 17 sites' Object_ID codes. These sites, along with the current EMS site, received a 0.5 mile buffer to show what communities are within range of the sites. It was found that the proposed locations covered far more high density communities than the current, single location.

Sources:

Lee County Florida GIS, EMS Station Locations, July 2, 2019; Lee County Florida GIS, Existing Landuse, June 28, 2019; Lee County Florida GIS, Major Roads, July 2, 2019; Lee County Florida GIS, Residential Neighborhoods and Commercial Areas, April 4, 2024; Lee County Florida GIS, Roads, November 12, 2019

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